Name.....

Signature……..**Candidates Personal Number**…….

545/1

CHEMISTRY.

Paper 1.

Aug /2025.

2 hours

CHEMISTRY DEPARTMENT

New Lower Secondary Chemistry Examination

Ugandan Certificate of Education

Internal mock

Paper 1

2 hours

INSTRUCTIONS:

This paper consists of **two** *sections* **A** *and* **B**. *It has* **six** *examination items.*

Section **A** *has* **one compulsory** items.

Section **B** *has* **two parts, I** and **II**. *Answer* **one** item from each part..

Answer **four** items in all.

Answers to section **A must** *be written in the spaces provided while those of section* **B must** *be written on the provided answer sheets.*

Any additional item(s) answered will **not** *be scored*

Where necessary use , C = 12, H = 1, O = 16, Na = 23

SECTION A

Answer **all** *items in this section in the spaces provided.*

Item 1

A company specializing in nuclear technology uses a radioactive isotope for different purposes . A sample of this isotope with an initial mass of 100 grams is being prepared for a test before use. The company's technical team has determined that the isotopes' nucleus **X**, (Atomic mass 60 and atomic number 27) transforms into a new daughter nucleus **Y**(Atomic mass 60 and atomic number 28) emitting particular radiations. For safety and regulatory compliance, the company must accurately track the isotope's decay. A measurement taken after 10.5 years shows that the mass of original isotope has decreased to 25 percent of its initial mass.

Task. As a Chemistry student,

(a) Name the type of radiations emitted.

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(b) Determine the half life of this radioactive isotope.

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(c) Suggest any industrial application of the radiations emitted.

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(d) What challenges are associated with the use of radiations emitted.?

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Task 2

An investor wishes to establish a small- scale chemical production facility using ethanoic acid as a primary raw material. One of the intended products is sodium ethanoate, a salt extensively used as a food preservative, to be prepared by reacting ethanoic acid and soda ash (sodium carbonate) as equation below.

$$2CH_3COOH(aq) + Na_2CO_3(s) \text{ ------- > } 2CH_3COONa(aq) + CO_2(g) + H_2O(l)$$

The investor requires to know the exact mass of soda ash that is needed to completely react with 0.25 kg of acid to produce the salt during p . This can guide him to minimize resources during scaling-up production and also need an overview of the environmental impact of the reaction. You have been invited for guidance

Task. As a learner of chemistry,

(a) explain the category of organic substance used to produce sodium ethanoate.

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(b) Suggest the suitability of organic raw material to its applications in every day life.

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(c) Calculate the mass of soda ash required in above reaction .

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(d) explain the environmental impact associated with the gaseous product.

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SECTION B

Part I

Attempt one item from this part

Item 3.

Due to the need to increase agricultural production, the government has cleared a local investor to establish ammonium nitrate fertilizer production industry in Jinja industrial area. However, the locals would like to know how the process of production will be carried out, how they have raised concern about how they will benefit from this investment and environment responsibility.

Task. As a chemistry student, make a write up you will use during the presentation.

Item 4.

The government through Uganda Investments Authority has cleared an investor to set up a cement production facility in one of the district of Uganda. However the locals are concerned about it's impact resulting from the process of production. You have been invited to address the locals in one of the sensitization workshops conducted at the sub county hall

Task. Write a write up presentation you will use during the presentation.

Part II.(*Attempt one item from this part.*)

Item 5

The Kampala city council Authority, KCCA is moving forward with a plan to partner with an investor to establish a biogas facility at kiteezi landfill. This investment is a crucial step towards cleaner city and energy sustainability. Recognizing the public environmental concern, KCCA officials have launched an education campaign to make the public informed about this megaproject. You have been invited to attend and address the public at a radio talkshow.

Task. As a chemistry prepare a short presentation you will deliver in the radio talk show.

Item 6

Upon the discovery of the rich crude oil reserves in the Albertine region, the locals raised environmental impact associated with its exploitation. They also lacked comprehensive knowledge of importance of crude oil. This has prompted the government to conduct a regional workshops to raise awareness amongst the public to be informed about crude oil . You have been invited to attend address the public.

Task. As a chemistry student, prepare a short message you will use to deliver in the workshop.

End.